GMC LANKA CREATIONS

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CHAPTER 1 INTRODUCTION

This Software Requirement Specification (SRS) document is prepared to provide a high-level overview of the GMC Lanka Machine and Product Ordering System. The system is designed to support the digital management of machine designs, customer orders, and service-related operations within the organization.

The system is intended to facilitate interactions between customers, administrators, and technicians through an integrated platform. It is designed to improve efficiency, enhance communication, and support better management of machine-related services.

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) document is to define and describe the functional and non-functional requirements of the GMC Lanka Machine and Product Ordering System.

This document is intended for use by project developers, the project supervisor, client representatives, and other stakeholders involved in the system development process. It is used to provide a clear understanding of system requirements, to guide the design and implementation process, and to ensure that the developed system satisfies the needs and expectations of the client.

This document also serves as a reference for how the system will operate and how interactions between different types of users, including customers, administrators, and technicians, will be handled within the system.

1.2 Scope

The GMC Lanka Machine and Product Ordering System is developed as a web-based platform to support the digital transformation of machine ordering and service management processes within GMC Lanka.

The system is intended to align with the overall business goals of improving operational efficiency, enhancing customer experience, and reducing manual workload in order processing and communication. A centralized platform is provided through which machine and product information can be managed and accessed efficiently.

The following objectives are intended to be achieved through the system

- A streamlined process for browsing machine and product designs is provided to customers.
- An efficient order placement and tracking mechanism is enabled.
- Communication between customers and administrators is improved through appointment booking and inquiry handling.
- Support for custom machine design requests is facilitated.
- Maintenance and service scheduling processes are organized and recorded digitally.

The following benefits are expected to be achieved

- Reduction in manual paperwork and human errors.
- Faster processing of customer orders and requests.
- Improved customer satisfaction through real-time tracking and communication.
- Better management of machine, product, and service data.

The system is designed to allow customers to view machine designs, place orders, request custom solutions, and track order status. Administrators are enabled to manage products, approve orders, generate invoices, and monitor system activities.

However, the system is limited to managing the digital processes related to ordering and service management. The physical manufacturing and delivery of machines are not included within the scope of this system.

1.3 Definitions, Acronyms and Abbreviations

Table 1.1 :Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirement Specification document that defines system requirements
Admin	System administrator responsible for managing system operations
Customer	A user who views machine designs and places orders
B2B	Business-to-Business transactions between companies
B2C	Business-to-Customer transactions between a business and individual customers

1.4 Document overview

This document is organized into several sections in which the requirements of the proposed system are described. The Introduction section is used to present the purpose, scope, and definitions related to the system. The Project Scope section is used to explain the overall objectives and system boundaries. The Stakeholders section is used to identify the individuals and organizations involved in or affected by the system.

The Functional Requirements and Non-Functional Requirements sections are used to describe the system features and quality attributes. The document also includes Business Rules, Constraints, and Assumptions, through which the operational conditions of the system are defined.

Further sections are presented in which Use Case Diagrams, Use Case Descriptions, and Activity Diagrams are used to illustrate system interactions. The External Interface Requirements and Data Requirements sections are used to explain how communication with users and other systems is handled.

Finally, sections on Alternative Solutions, Feasibility Study, Risk Analysis, and References are included to support the proposed system and its development.

CHAPTER 2 PROJECT SCOPE

The GMC Lanka Machine and Product Ordering System is developed as a web-based platform to support the management and ordering of industrial machines and product designs offered by GMC Lanka.

Customers are allowed to browse available machine designs and product details through an online interface. Orders can be placed, order progress can be tracked, custom machine designs can be requested, and appointments can be scheduled with the administrator to discuss machine requirements.

Support is provided to the administrator for managing machine and product information, reviewing and approving customer orders, handling user accounts, and generating invoices or quotations for customers. Additionally, maintenance and service scheduling features are included to manage repair and service history of machines.

The efficiency of machine ordering, customer communication, and service management at GMC Lanka is improved by replacing manual processes with an integrated digital platform.

However, the physical manufacturing process of machines is not included in the system. Only the information and workflow related to machine design viewing, ordering, and service management are managed.

CHAPTER 3 STAKEHOLDERS

Stakeholders are individuals or organizations that are involved in the development, use, or management of the system.

Table 3.1: Stakeholders and Their Roles

Stakeholder	Description	Role in the System
GMC Lanka Management	The organization requesting the system	Provides requirements and approves the system
Administrator	Staff responsible for managing the system	Manages machines, users, orders, and services
Customers	Individuals or businesses purchasing machines	Browse machines, place orders, request services
Technicians	Technical staff responsible for machine maintenance	Manage repair records and service scheduling
Development Team	Student developers creating the system	Design, develop, and maintain the system
Project Supervisor	Academic supervisor from the university	Guides to the project and evaluates progress

CHAPTER 4 FUNCTIONAL REQUIREMENTS

4.1 User management

Table 4.1 : User management requirements

Id	Description	Priority	Source	Status
FR001	The system shall allow users to register with a unique email address and password	High	Client	Approved
FR002	The system shall support role-based access control with roles: Admin and Customer.	High	Client	Approved
FR003	The system shall authenticate users via secure login with email and password.	High	Client	Approved
FR004	The admin shall be able to create, update, and deactivate customer accounts.	High	Admin	Approved

4.2 Product management

Table 4.2 : Product management requirements

Id	Description	Priority	Source	Status
FR005	The admin shall be able to add, update, and delete machine and wood craft product listings.	High	Admin	Approved
FR006	The system shall allow the admin to upload product images, specifications, and manuals.	High	Admin	Approved
FR007	The system shall allow products to be categorised by type (e.g., machines, wood crafts).	Medium	Admin	Approved

FR008	Customers shall be able to browse and search the product catalogue	High	Customer	Approved
FR009	Customers shall be able to submit a custom machine or wood design request with their requirements.	High	Client	Approved
FR010	The admin shall be able to review and respond to custom design requests with design suggestions.	High	Admin	Approved
FR011	Once a custom design is agreed upon, the customer shall be able to place an order for it.	High	Client	Approved

4.3 Order management

Table 4.3 : Order management requirements

Id	Description	Priority	Source	Status
FR012	Customers shall be able to place new B2C orders for catalogue products	High	Customer	Approved
FR013	The system shall support B2B order placement with bulk quantity handling.	High	Client	Approved
FR014	Customers shall be able to track the real-time status of their orders.	High	Customer	Approved
FR015	The admin shall be able to approve, update, or cancel orders.	High	Admin	Approved

FR016	The system shall automatically generate invoices and quotations for orders.	High	Client	Approved

4.4 Payment management

Table 4.4 : Payment management requirements

Id	Description	Priority	Source	Status
FR017	The system shall support multiple payment modes: cash, card, and bank transfer.	High	Client	Approved
FR018	The system shall integrate with Stripe for secure online card payment processing.	Medium	Customer	Approved
FR019	The system shall generate and send payment receipts to customers upon successful payment.	Medium	Customer	Approved

4.5 Maintenance management

Table 4.5 : Maintenance management requirements

Id	Description	Priority	Source	Status
FR020	The admin shall be able to approve, update, or cancel orders.	High	Admin	Approved

FR021	The system shall maintain a repair and service history log for each machine.	Medium	Customer	Approved
FR022	Customers shall be able to submit service or repair requests for purchased machines.	Medium	Customer	Approved

4.6 Booking management

Table 4.6 : Booking management requirements

Id	Description	Priority	Source	Status
FR023	Customers shall be able to request an appointment with the admin to discuss machine or design details.	Medium	Admin	Approved
FR024	Customers shall be able to request an appointment with the admin to discuss machine or design details.	Medium	Admin	Approved

4.7 Notification management

Table 4.7 : Notification management requirements

Id	Description	Priority	Source	Status
FR025	The system shall display real-time desktop alerts for upcoming appointments, delivery dates, and new bookings	High	Admin	Proposed

FR026	The system shall notify the admin of low stock levels for machine parts.	Medium	Admin	Proposed
FR027	The system shall notify customers of pending bills and order status changes	Medium	Customer	Proposed

4.8 Support and feedback management

Table 4.8 : Support and feedback management requirements

Id	Description	Priority	Source	Status
FR028	Customers shall be able to submit complaints or queries through the system.	Medium	Customer	Proposed
FR029	The system shall allow customers to provide ratings and feedback on products and services.	Low	Customer	Proposed
FR030	The admin shall be able to track and update the resolution status of customer complaints.	Medium	Admin	Proposed

CHAPTER 5 NON-FUNCTIONAL REQUIREMENTS

5.1 Performance

Table 5.1 : Performance

ID	Category	Measurable Acceptance Criterion	Priority	Source	Status
NFR001	Performance	95% of search queries shall return results within 2 seconds under normal load.	High	Client	Proposed
NFR002	Performance	The catalogue page shall load within 3 seconds for up to 50 concurrent users.	High	Client	Proposed

5.2 Reliability

Table 5.2 : Reliability

ID	Category	Measurable Acceptance Criterion	Priority	Source	Status
NFR003	Reliability	System uptime shall be at least 99% during business hours (8 AM – 8 PM).	High	Client	Proposed
NFR004	Reliability	The system shall restore full functionality within 4 hours of any unplanned outage.	High	Development team	Proposed
NFR005	Reliability	0% data loss shall be recorded during synchronization events, verified by automated sync integrity tests across 100% of test	High	Development team	Proposed

		cases.			

5.3 Usability

Table 5.3 : Usability

ID	Category	Measurable Acceptance Criterion	Priority	Source	Status
NFR007	Usability	A new admin user shall be able to complete core tasks within 30 minutes without training.	Medium	Client	Approved
NFR008	Usability	All validation error messages shall be human-readable and displayed within 1 second of user action.	Medium	Client	Approved

5.4 Security

Table 5.4 : Security

ID	Category	Measurable Acceptance Criterion	Priority	Source	Status
NFR009	Security	Passwords shall be hashed using bcrypt with a minimum cost factor of 10 before storage	High	Development team	Approved
NFR010	Security	All payment data shall be transmitted over HTTPS/TLS 1.2 or higher and processed via Stripe PCI-DSS compliant API.	High	Client	Approved

NFR011	Security	No customer user shall be able to access admin-only pages; verified by 100% of access control test cases passing.	High	Client	Approved

5.5 Scalability

Table 5.5 : Scalability

ID	Category	Measurable Acceptance Criterion	Priority	Source	Status
NFR012	Scalability	The system shall support at least 100 concurrent users without performance degradation.	High	Development team	Approved
NFR013	Scalability	New modules, such as AI-powered design recommendations or advanced reporting, shall be integrable without modifying more than 20% of the existing codebase, verified by architectural review.	High	Development team	Approved

5.6 Maintainability

Table 5.6 : Maintainability

ID	Category	Measurable Acceptance Criterion	Priority	Source	Status
NFR014	Maintainability	Code coverage by automated tests shall be at least 70% of core modules.	High	Development team	Proposed

NFR015	Maintainability	100% of system updates and patches shall be deployable with zero downtime, using rolling deployment or blue-green deployment strategies, verified during each release cycle.	High	Development team	Proposed
NFR016	Maintainability	All system logs — including order activity, login attempts, payment events, and error records — shall be retained for a minimum of 12 months and shall be retrievable within 30 seconds of a query.	High	Development team	Proposed

5.7 Portability

Table 5.7 : Portability

ID	Category	Measurable Acceptance Criterion	Priority	Source	Status
NFR017	Portability	The system shall function correctly on Chrome (v110+), Firefox (v110+), and Edge (v110+).	High	Client	Proposed

NFR018	Portability	The system shall be deployable to both AWS cloud-hosted and on-premises server environments without requiring changes to more than 10% of the configuration or codebase, verified by a successful migration test prior to final deployment.	High	Developme nt team	Proposed

CHAPTER 6 BUSINESS RULES

- An order may only be placed if the customer has a registered and verified account
- Custom design orders require admin approval before progressing to production.
- A machine maintenance service must be logged before the machine is marked as serviced.
- Invoices shall be automatically generated upon order confirmation and cannot be deleted.
- An appointment slot may only be booked if it is marked as available by the admin.
- Payments must be confirmed before an order status is updated to 'Processing'.
- Low stock alerts shall be triggered when parts inventory falls below a configurable threshold.

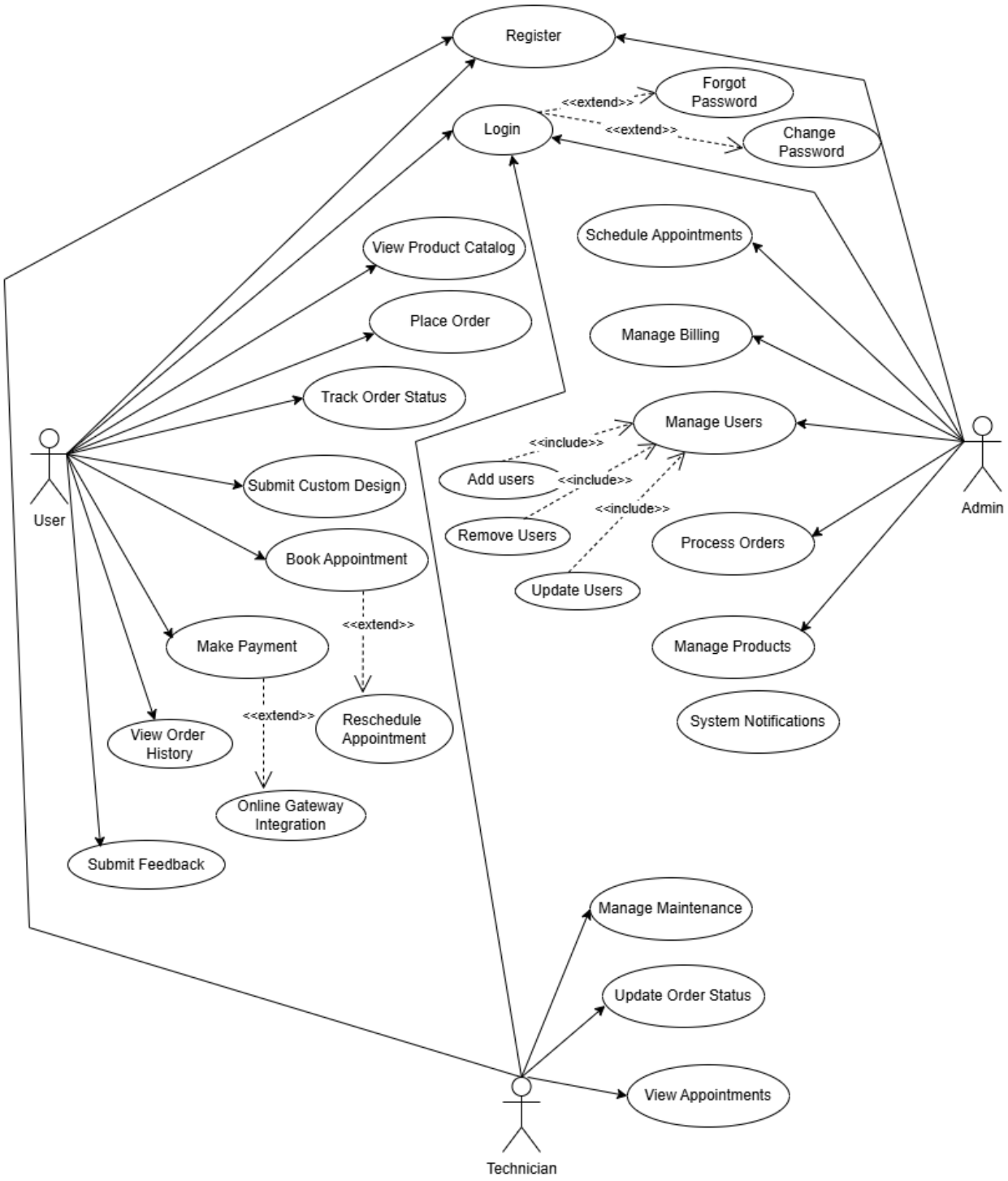
CHAPTER 7 CONSTRAINTS

- The system must be developed using React.js (frontend), and a compatible backend framework, MySQL database, and AWS cloud hosting.
- Online payments must be processed exclusively through the Stripe payment gateway.
- The system is intended for desktop use and need not be optimised for mobile devices in this phase.
- The system must be completed and delivered within the academic project timeline.
- All data must comply with applicable data protection regulations.

CHAPTER 8 ASSUMPTIONS

- The client and admin users will have consistent access to a stable internet connection.
- The client will provide required product images, machine specifications, and content for catalogue setup.
- The Stripe payment gateway will remain operational and accessible in Sri Lanka.
- The client will manage AWS infrastructure costs independently after deployment.
- All users are assumed to have basic computer literacy sufficient to use a web browser.

CHAPTER 9 USE CASE DIAGRAMS



CHAPTER 10 USE CASE DESCRIPTIONS

Table 10.1: Use case description of User Register

Use Case ID	UC001
Use Case Name	User Register
Description	Allows a new customer to register with the system by providing personal details to create an account.
Actor	User
Pre-Conditions	The user is not already registered in the system. The user has access to the registration page.
Post Conditions	The user account is created successfully. Users can log into the system.
Main Flow	1. The user navigates to the Registration/Sign-up page. 2. The system prompts the user to enter their details (name, email, password, etc.). 3. The system validates the input data (e.g., password strength, email format). 4. Upon successful validation, the system creates an account for the user. 5. The system sends a confirmation email or message to verify the registration.

Alternative Flows	Invalid Credentials: If the input data is invalid (e.g., weak password, incorrect email format), the system prompts the user to correct the information. Existing Account: If the email is already associated with an existing account, the system notifies the user to log in or reset their password.

Table 10.2: Use case description of User Login

Use Case ID	UC002
Use Case Name	Login
Description	Allows a new customer to register with the system by providing personal details to create an account.
Actor	User
Pre-Conditions	Users must have an active account in the system. The user has access to the login page.
Post Conditions	Users are authenticated and redirected to their respective dashboard.

Main Flow	1. The user navigates to the login page. 2. The system prompts the user to enter their username/email and password. 3. The system validates the credentials. 4. Upon successful validation, the user is logged in and redirected to their dashboard.
Alternative Flows	Invalid Credentials: If the entered credentials are invalid, the system displays an error message and prompts the user to re-enter their username and password. Account Locked: After multiple failed attempts, the account may be temporarily locked for security.
Extensions	• Forgot Password: If the user forgets their password, they can opt to recover it using the "Forgot Password" feature. • Change Password: The user may also update their password after successfully logging in.

Table 10.3: Use case description of User Forgot Password

Use Case ID	UC003
Use Case Name	Forgot Password
Description	The user requests a password reset by providing a registered email address to receive reset instructions.

Actor	User
Pre-Conditions	The user is registered and has access to registered email. The user is on the login page.
Post Conditions	Password reset link/instructions are sent to the user's email.
Main Flow	1. The user clicks on the "Forgot Password" link on the login page. 2. The system prompts users to enter a registered email address. 3. The user enters an email and submits. 4. The system validates that the email exists in the database. 5. The system sends password reset instructions to the provided email. 6. The system displays a confirmation message to the user.
Alternative Flows	Email Not Found: If the entered email is not registered, the system notifies the user and prompts them to try again or register. Email Delivery Failure: If the reset email cannot be sent, the system notifies the user and provides alternative support contact.

Table 10.4: Use case description of User Change Password

Use Case ID	UC004
Use Case Name	Change Password

Description	Users change their current password to a new password after successful authentication.
Actor	User
Pre-Conditions	The user is logged into the system.
Post Conditions	The user's password is updated successfully in the system.
Main Flow	1. User navigates to profile settings or account management page. 2. The user selects the "Change Password" option. 3. The system prompts users to enter their current password and new password. 4. The user enters the required information. 5. System validates current password and new password strength. 6. Upon validation, the system updates the password. 7. The system displays a success message and may require re-login.
Alternative Flows	Incorrect Current Password: If current password is incorrect, system displays error and prompts user to try again. Weak New Password: If the new password does not meet security requirements, the system prompts the user to choose a stronger password.

Table 10.5: Use case description of User View Product Catalog

Use Case ID	UC005
Use Case Name	View Product Catalog
Description	Users browse the available CNC-carved products, machines, and wood designs categorized by type.
Actor	Users
Pre-Conditions	The user is logged into the system. The product catalog is populated with data.
Post Conditions	Product catalog is displayed with images, specifications, and details.
Main Flow	1. The user navigates to the "Products" or "Catalog" section. 2. The system retrieves product listings from the database. 3. The system displays products categorized by type (CNC, Cutting, Grinding, Drilling). 4. Users can filter or sort products based on preferences. 5. Users can select individual products to

	view detailed information including images, manuals, and specifications.
Alternative Flows	No Products Available: If no products are available, the system displays a "No products found" message. Category Empty: If a selected category has no products, the system notifies the user and suggests browsing other categories.

Table 10.6: Use case description of User Submit Custom Design

Use Case ID	UC006
Use Case Name	Submit Custom Design
Description	Users upload custom design requirements and files for a personalized CNC-carved product.
Actor	User
Pre-Conditions	The user is logged in and has design files ready for upload.
Post Conditions	A custom design request is submitted and sent to the admin for review.

Main Flow	1. The user navigates to the "Custom Design" or "Submit Design" section. 2. The system displays a form for design submission. 3. The user enters design details, specifications, and requirements. 4. Users upload design files (images, CAD files, etc.). 5. The user submits the form. 6. The system validates the uploaded files and enters information. 7. The system saves the custom design request and notifies the admin. 8. The system displays a confirmation message to the user with a request ID.
Alternative Flows	Invalid File Format: If uploaded files are in unsupported formats, the system rejects them and prompts users to upload acceptable formats. File Size Exceeded: If files exceed size limits, the system notifies the user and requests smaller files. Missing Required Fields: If required fields are incomplete, the system highlights missing information and prevents submission.

Table 10.7: Use case description of User Book Appointment

Use Case ID	UC007
Use Case Name	Book Appointment
Description	The user schedules an appointment with the admin to clarify machine or design details.
Actor	User
Pre-Conditions	The user is logged into the system. Admin availability is configured.
Post Conditions	Appointment request is created and pending admin approval.
Main Flow	1. The user navigates to the "Book Appointment" section. 2. The system displays available date and time slots. 3. The user selects the preferred date and time. 4. User enters appointment purpose and any additional notes. 5. The user submits the appointment request. 6. The system saves the appointment with "Pending" status. 7. The system sends confirmation to the user and notification to the admin. 8. The system displays appointment details to users.

Alternative Flows	No Available Slots: If no slots are available, the system suggests alternative dates or notifies the user to check later. Duplicate Booking: System prevents double-booking of same time slot for same user.
Extensions	• Reschedule Appointment: Users can modify an existing appointment.

Table 10.8: Use case description of User Research Appointment

Use Case ID	UC008
Use Case Name	Research Appointment
Description	The user modifies an existing appointment to a different date or time.
Actor	User
Pre-Conditions	The user has an existing appointment that is not yet completed or cancelled.
Post Conditions	Appointment is updated with new schedule details.

Main Flow	1. The user navigates to the "My Appointments" section. 2. The system displays a list of user's appointments. 3. The user selects an appointment and chooses "Reschedule". 4. The system displays available date and time slots. 5. The user selects a new date and time. 6. The user confirms the reschedule request. 7. The system updates the appointment and notifies the admin. 8. The system displays updated appointment details to users.
Alternative Flows	Cancellation Period Expired: If rescheduling is not allowed close to appointment time, the system prevents action and notifies the user. Admin Disapproval: If admin rejects reschedule request, system notifies user and maintains original appointment.

Table 10.9: Use case description of User Place Order

Use Case ID	UC009
Use Case Name	Place Order

Description	Users place a new order for selected products from the catalog (B2B & B2C).
Actor	User
Pre-Conditions	The user has selected products and added them to cart. The user is logged in.
Post Conditions	Order is created and sent to the admin for processing.
Main Flow	1. The user navigates to the cart with selected items. 2. User reviews order summary including quantities and prices. 3. The user proceeds to checkout. 4. The system prompts users to confirm shipping/billing details. 5. The user confirms the order details. 6. The system creates order with "Pending" status. 7. The system generates order confirmation and sends it to the user. 8. The system notifies the admin of the new order.
Alternative Flows	Empty Cart: If a user attempts to place an order with an empty cart, the system redirects to the product catalog. Out of Stock: If any item is out of stock

	during checkout, the system notifies users and suggests alternatives or removal.
Extensions	• Make Payment: User proceeds to payment after order placement.

Table 10.10: Use case description of User Make Payment

Use Case ID	UC010
Use Case Name	Make Payment
Description	The user completes payment for the placed order using available payment modes (cash, card, bank transfer).
Actor	User
Pre-Conditions	Order is placed and payment amount is calculated. The user is authenticated.
Post Conditions	Payment is processed and receipt is generated.

Main Flow	1. After order confirmation, the user selects the payment method. 2. The system displays payment options (cash, card, bank transfer). 3. The user selects a preferred payment method. 4. If online payment, the system redirects to the payment gateway. 5. The user completes payment information and submits. 6. The system verifies payment success with gateway. 7. System updates order payment status. 8. The system generates and displays auto-generated receipts. 9. The system sends payment confirmation to the user.
Alternative Flows	Payment Failed: If payment fails, the system notifies the user and provides options to retry or choose a different method. Cash on Delivery: If the user selects cash, the system notes the payment method and updates order accordingly. Gateway Timeout: If payment gateway times out, the system checks payment status and updates users accordingly.

Includes	• Online Gateway Integration: System integrates with external payment gateways for online transactions.

Table 10.11: Use case description of User Track Order Status

Use Case ID	UC011
Use Case Name	Track Order Status
Description	Users view the real-time status of their order (Pending, In Progress, Completed, Delivered).
Actor	User
Pre-Conditions	The user has placed an order. The user is logged in.
Post Conditions	Current order status is displayed.
Main Flow	1. The user navigates to the "My Orders" or "Order History" section. 2. The system displays a list of the user's orders. 3. The user selects a specific order to view details. 4. The system displays order details including current status.

	5. The system shows a timeline of status updates if available.
Alternative Flows	Order Not Found: If the selected order does not exist or the user doesn't have permission, the system displays an error message. Status Unavailable: If status information is temporarily unavailable, the system notifies the user and suggests checking later.

Table 10.12: Use case description of User View Order History

Use Case ID	UC012
Use Case Name	View Order History
Description	The user accesses a list of all past orders with details and statuses.
Actor	User
Pre-Conditions	The user is logged in and has previous orders in the system.

Post Conditions	Order history is displayed with relevant details.
Main Flow	1. The user navigates to "Order History" or "My Account" section. 2. The system retrieves all orders associated with the user. 3. The system displays orders sorted by date (most recent first). 4. Users can filter or search through order history. 5. Users can select individual orders to view complete details.
Alternative Flows	No Order History: If the user has no previous orders, the system displays a message indicating no orders found and provides a link to the product catalog.

Table 10.13: Use case description of User View Order History

Use Case ID	UC013
Use Case Name	Submit Feedback
Description	Users provide feedback, complaints, or ratings for products and services received.
Actor	User

Pre-Conditions	The user has received an order or used a service. The user is logged in.
Post Conditions	Feedback is recorded in the system for admin review with tracking capability.
Main Flow	1. The user navigates to the "Feedback" or "Support" section. 2. The user selects order or service to provide feedback on. 3. The system displays feedback form with rating options. 4. The user enters feedback/complaint details and rating. 5. The user submits the feedback. 6. The system records feedback with a unique tracking ID. 7. System notifies admin of new feedback/complaint. 8. The system displays confirmation to the user with tracking ID for resolution follow-up.
Alternative Flows	Anonymous Feedback: System may allow feedback without order reference for general suggestions. Urgent Complaint: System may flag certain feedback as urgent based on keywords for immediate admin attention.

Table 10.14: Use case description of Admin Manage Users

Use Case ID	UC014
Use Case Name	Manage Users
Description	Admin oversees user accounts including adding, updating, and removing users from the system with role-based access control.
Actor	Admin
Pre-Conditions	Admin is logged in with appropriate privileges.
Post Conditions	User records are updated in the system.
Main Flow	1. Admin navigates to the "User Management" section. 2. The system displays a list of all users with their roles (Admin, Customers, Technicians). 3. Admin can search, filter, and view user details. 4. Admin selects actions (Add, Update, Remove) on users. 5. The system processes the requested action. 6. System confirms successful operation and updates user list.
Alternative Flows	Permission Denied: If admin attempts actions beyond their system displays access denied message. User Not Found: If trying to

	update/remove a non-existent user, the system displays error.
Includes	• Add Users: Admin creates new user accounts. • Remove Users: Admin deactivates or removes existing users. • Update Users: Admin modifies user details and permissions.

Table 10.15: Use case description of Admin Add Users

Use Case ID	UC015
Use Case Name	Add Users
Description	Admin creates new user accounts for customers, technicians, or other staff.
Actor	Admin
Pre-Conditions	Admin is logged in. New user details are available.
Post Conditions	A new user account is created with the appropriate role.

Main Flow	1. Admin selects the "Add User" option from user management. 2. The system displays user creation forms. 3. Admin enters user details (name, email, role, etc.). 4. Admin sets temporary password or triggers invitation email. 5. Admin submits the form. 6. The system validates and creates the user account. 7. The system sends account creation notification to new users. 8. System confirms success to admin.
Alternative Flows	Duplicate Email: If email already exists, the system notifies admin and prevents duplicate account creation. Invalid Data: If required fields are missing or invalid, the system highlights errors and prevents submission.

Table 10.16: Use case description of Admin Remove Users

Use Case ID	UC016
Use Case Name	Remove Users
Description	Admin deactivates or removes existing user accounts from the system.

Actor	Admin
Pre-Conditions	Admin is logged in and the target user exists.
Post Conditions	The user account is deactivated or removed from the active system.
Main Flow	1. Admin selects user from user list. 2. Admin chooses the "Remove User" option. 3. The system displays confirmation dialog with warnings about consequences. 4. Admin confirms removal. 5. System deactivates or removes the user account. 6. The system logs the action and notifies relevant parties if configured. 7. System confirms removal to admin.
Alternative Flows	Self-Removal Prevention: System prevents admin from removing their own account. User Has Active Orders: System warns admin if user has active orders and suggests handling them before removal.

Table 10.17: Use case description of Admin Update Users

Use Case ID	UC017
Use Case Name	Update Users

Description	Admin modifies user details, roles, and permissions.
Actor	Admin
Pre-Conditions	Admin is logged in and the target user exists.
Post Conditions	User information is updated successfully.
Main Flow	1. Admin selects user from user list. 2. Admin chooses "Edit" or "Update" option. 3. The system displays current user details in editable form. 4. Admin modifies necessary fields (name, email, role, permissions). 5. Admin submits changes. 6. The system validates updated information. 7. The system saves changes and logs the update. 8. System confirms successful update to admin.
Alternative Flows	Invalid Data: If updated data is invalid, system rejects changes and prompts for correction. Permission Change: If changing roles, the system may require additional confirmation for security.

Table 10.18: Use case description of Admin Manage Products

Use Case ID	UC018
Use Case Name	Manage Products
Description	Admin adds, updates, or deletes machine/wood design details including images, manuals, and specifications, categorized by type.
Actor	Admin
Pre-Conditions	Admin is logged in. Product information is available for addition/update.
Post Conditions	Product catalog is updated with new/changed information.
Main Flow	1. Admin navigates to "Product Management" section. 2. The system displays current products by category. 3. Admin selects action (Add, Edit, Delete). 4. For add/edit, the system displays product form. 5. Admin enters/updates product details: name, description, category, price, images, manuals, specifications. 6. Admin uploads media files if applicable. 7. Admin submits changes. 8. The system validates and saves product information.

	9. System confirms success to admin and updates catalog.
Alternative Flows	Duplicate Product: If a product with the same name exists, the system warns admin. Invalid Category: If category doesn't exist, system prompts admin to select valid category. File Upload Error: If images/manuals fail to upload, system notifies admin and allows retry.

Table 10.19: Use case description of Admin Process Orders

Use Case ID	UC019
Use Case Name	Process Orders

Description	Admin reviews and processes incoming customer orders.
Actor	Admin
Pre-Conditions	Orders have been placed by customers. Admin is logged in.
Post Conditions	Orders are reviewed and processing begins.
Main Flow	1. Admin navigates to the "Order Management" section. 2. The system displays a list of new orders requiring attention. 3. Admin selects an order to review details. 4. Admin verifies order items, customer information, and payment status. 5. Admin initiates order processing. 6. System updates order status to "In Progress". 7. The system notifies the customer that the order is being processed.
Alternative Flows	Suspicious Order: If order appears fraudulent, admin can flag for review or contact customer. Incomplete Information: If order lacks required details, admin can contact customer for clarification.
Extensions	• Update Order Status: Admin can update status throughout the fulfillment process.

Table 10.20: Use case description of Admin Update Order Status

Use Case ID	UC020
Use Case Name	Update Order Status
Description	Admin changes the status of orders through the fulfillment lifecycle (Pending, In Progress, Completed, Delivered).
Actor	Admin
Pre-Conditions	Orders exist in the system. Admin is logged in.
Post Conditions	Order status is updated and visible to customers with appropriate notifications.
Main Flow	1. Admin navigates to order details. 2. Admin selects current status and chooses new status from available options. 3. Admin may add notes or comments about the status change. 4. Admin confirms status update. 5. System updates order status. 6. The system triggers notification to customers about status change. 7. System logs the status change with timestamp and admin ID.

Alternative Flows	Invalid Status Transition: If attempting to move to invalid status (e.g., from Pending to Delivered), the system prevents action and displays allowed transitions. Bulk Update: Admin may be able to update status for multiple orders simultaneously.

Table 10.21: Use case description of Admin Manage Billing

Use Case ID	UC021
Use Case Name	Manage Billing
Description	Admin handles billing operations including generating invoices and quotations.
Actor	Admin
Pre-Conditions	Orders are processed or quotations are requested. Admin is logged in.
Post Conditions	Invoices and quotations are generated and sent to customers.

Main Flow	1. Admin navigates to the "Billing" or "Invoicing" section. 2. Admin selects order or customer for billing. 3. The system displays order details and calculated amounts. 4. Admin reviews and adjusts if necessary (discounts, taxes). 5. Admin generates invoice or quotation. 6. The system creates formal documents with unique numbers. 7. The system offers options to download, print, or email documents. 8. Admin sends documents to customers.
Alternative Flows	Recurring Billing: For B2B customers, the system may support recurring invoice generation. Manual Adjustment: If system calculations need adjustment, admin can manually modify before finalizing.

Table 10.22: Use case description of Admin View Appointments

Use Case ID	UC022
Use Case Name	View Appointments
Description	Admin views all scheduled appointments from customers and can approve, schedule, or reschedule them.

Actor	Admin
Pre-Conditions	Appointments have been booked by users. Admin is logged in.
Post Conditions	The appointment list is displayed and the admin can manage them.
Main Flow	1. Admin navigates to the "Appointment Management" section. 2. The system displays a calendar or list view of all appointments. 3. Admin can filter by date, status, or customer. 4. Admin selects an appointment to view details. 5. Admin can approve, reschedule, or cancel appointments. 6. The system updates appointment status and notifies customers.
Alternative Flows	Conflict Detection: If approving/rescheduling creates conflicts, the system warns admin. Overdue Appointments: System highlights appointments that are overdue or require immediate attention.

Table 10.23: Use case description of Admin Manage Maintenance

Use Case ID	UC023

Use Case Name	Manage Maintenance
Description	Admin schedules preventive maintenance for machines and logs repair/service history.
Actor	Admin
Pre-Conditions	Machines are registered in the system. Admin is logged in.
Post Conditions	Maintenance schedules are created and service history is logged.
Main Flow	1. Admin navigates to the "Maintenance Management" section. 2. Admin selects machines requiring maintenance. 3. Admin schedules preventive maintenance with date and tasks. 4. The system records the schedule and assigns to technicians if applicable. 5. After maintenance completion, the admin logs service details and history. 6. System updates machine maintenance records.
Alternative Flows	Recurring Maintenance: System may support recurring maintenance schedules with automatic reminders. Emergency Maintenance: Admin can log unscheduled/emergency maintenance separately.

Table 10.24: Use case description of Admin System Notifications

Use Case ID	UC024
Use Case Name	System Notifications
Description	Admin receives real-time desktop alerts for low stock, pending bills, and upcoming bookings.
Actor	Admin
Pre-Conditions	System events trigger notification conditions. Admin is logged in.
Post Conditions	Alerts are displayed on the admin dashboard or via desktop notifications.
Main Flow	1. System monitors predefined conditions (stock levels, payment due dates, appointment times). 2. When conditions are met, the system generates an alert. 3. Alert appears on admin dashboard notification center. 4. Desktop notification may pop up if enabled. 5. Admin views and acknowledges alerts. 6. System records acknowledgment and may clear alert after action.
Alternative Flows	Alert Dismissal: Admin can dismiss non-critical alerts.

	Escalation: Critical alerts may escalate if not addressed within a time frame.

Table 10.25: Use case description of Technician Login

Use Case ID	UC025
Use Case Name	Login
Description	Technician logs into the system using valid credentials with role-based access.
Actor	Technician
Pre-Conditions	Technician account is created by admin. The technician has valid credentials.
Post Conditions	Technician is authenticated and granted access to technician-specific dashboard and features.
Main Flow	1. Technician navigates to the login page. 2. Technician enters username/email and password. 3. The system validates credentials and roles. 4. Upon successful validation, technician is redirected to technician dashboard.
Alternative Flows	Invalid Credentials: System displays error message and prompts to retry. Account Inactive: If technician account is

	deactivated, system displays appropriate message.

Table 10.26: Use case description of Technician View Appointments

Use Case ID	UC026
Use Case Name	View Appointments
Description	Technician views assigned appointments and maintenance schedules.
Actor	Technician
Pre-Conditions	The technician is logged in. Appointments or maintenance tasks are assigned.
Post Conditions	Appointment details are displayed for technicians.

Main Flow	1. The technician navigates to "My Appointments" or "Assigned Tasks" section. 2. The system displays a list of appointments and maintenance tasks assigned to the technician. 3. Technicians can view details of each appointment including customer, date, time, and purpose. 4. The system may show a calendar view of scheduled tasks.
Alternative Flows	No Assignments: If no appointments are assigned, the system displays appropriate messages.

Table 10.27: Use case description of Technician Manage Maintenance

Use Case ID	UC027
Use Case Name	Manage Maintenance
Description	Technician performs maintenance tasks and updates service logs and history.
Actor	Technician
Pre-Conditions	Maintenance tasks are assigned. Technician is logged in.
Post Conditions	Service history is updated with completed maintenance records.

Main Flow	1. Technician selects assigned maintenance tasks. 2. The system displays task details including machine, required actions, and instructions. 3. Technician performs maintenance. 4. Technician logs completed actions, observations, and any issues encountered. 5. Technician updates task status to "Completed". 6. System records service history and notifies admin of completion.
Alternative Flows	Issue Encountered: If technician encounters problems requiring additional attention, they can flag tasks for admin review. Parts Required: Technicians can request parts or additional resources through the system.

CHAPTER 11 ACTIVITY DIAGRAMS

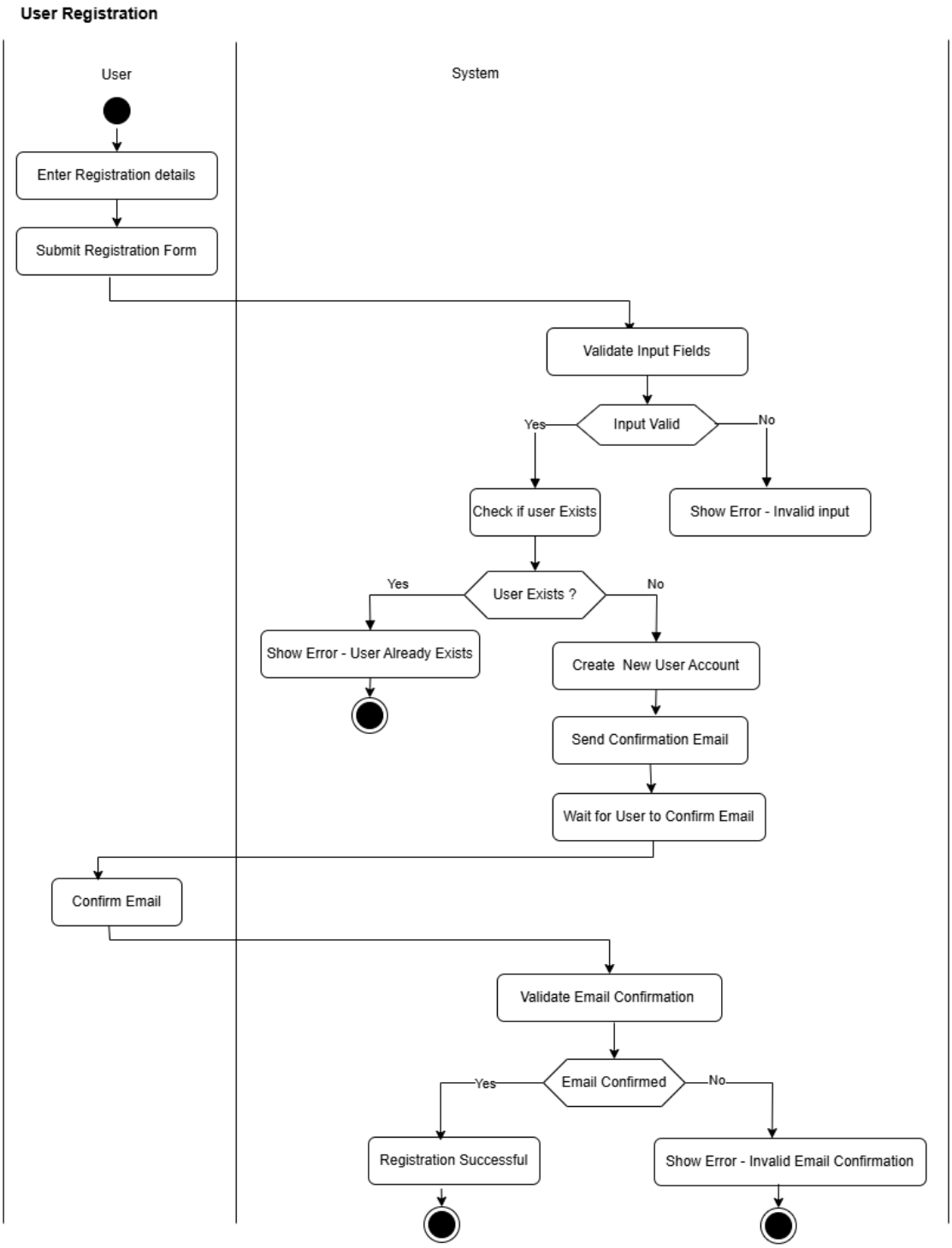


Figure 11.1: Activity Diagram — User Registration

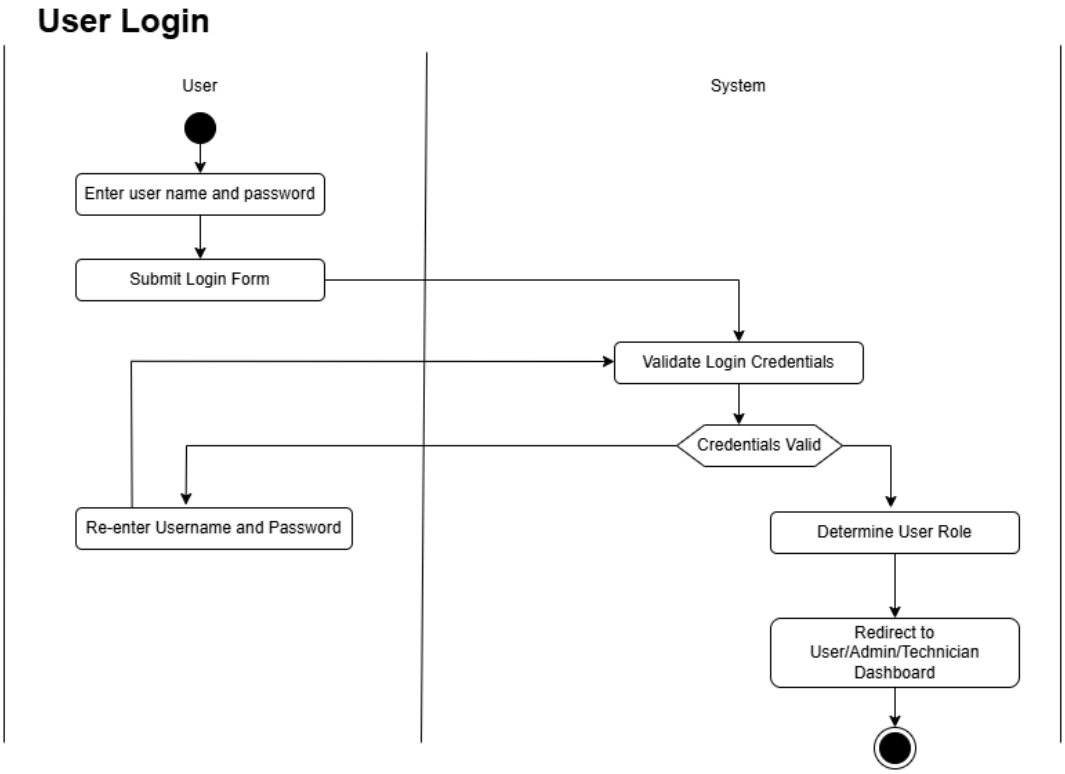


Figure 11.2: Activity Diagram — User login

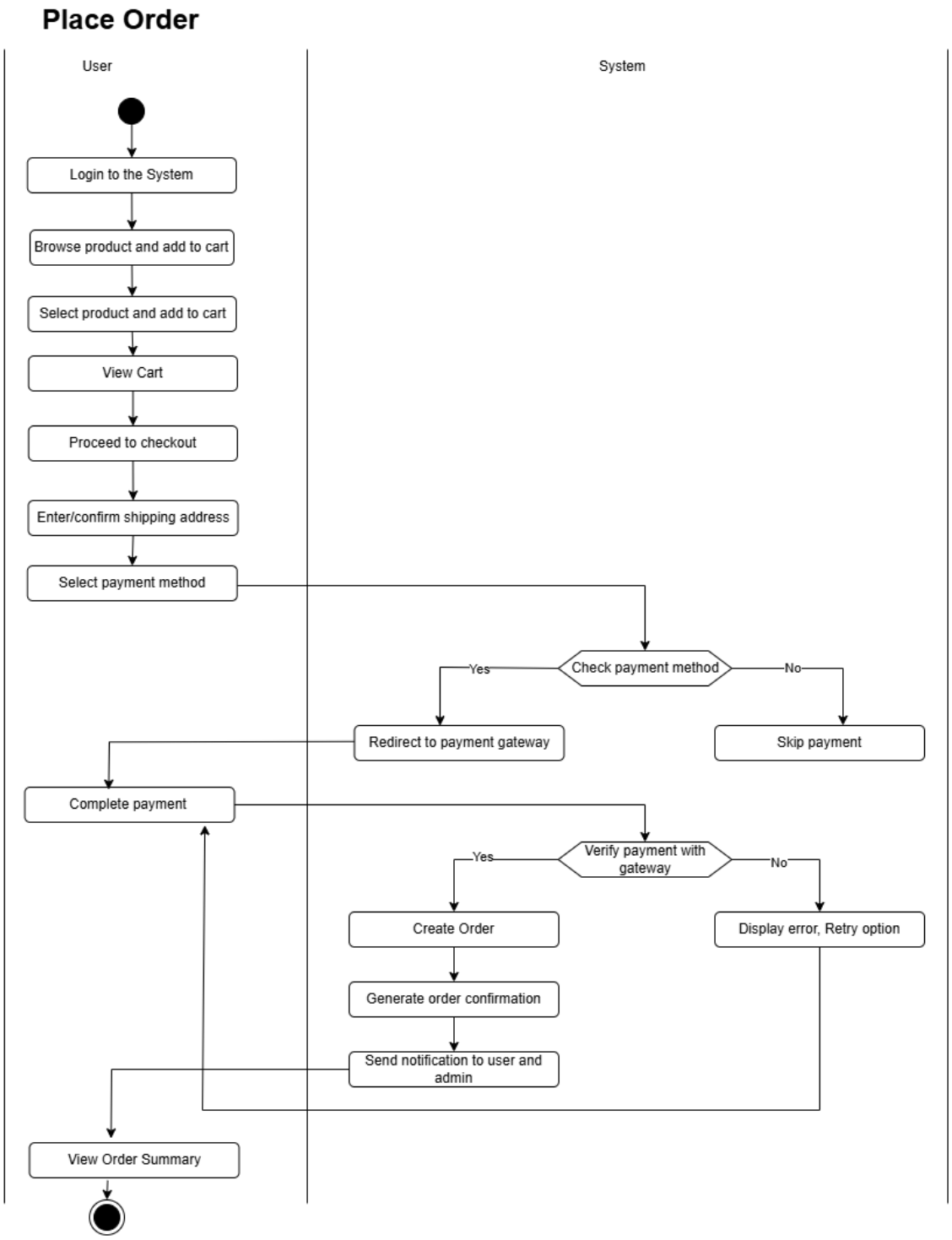


Figure 11.3: Activity Diagram — Place order

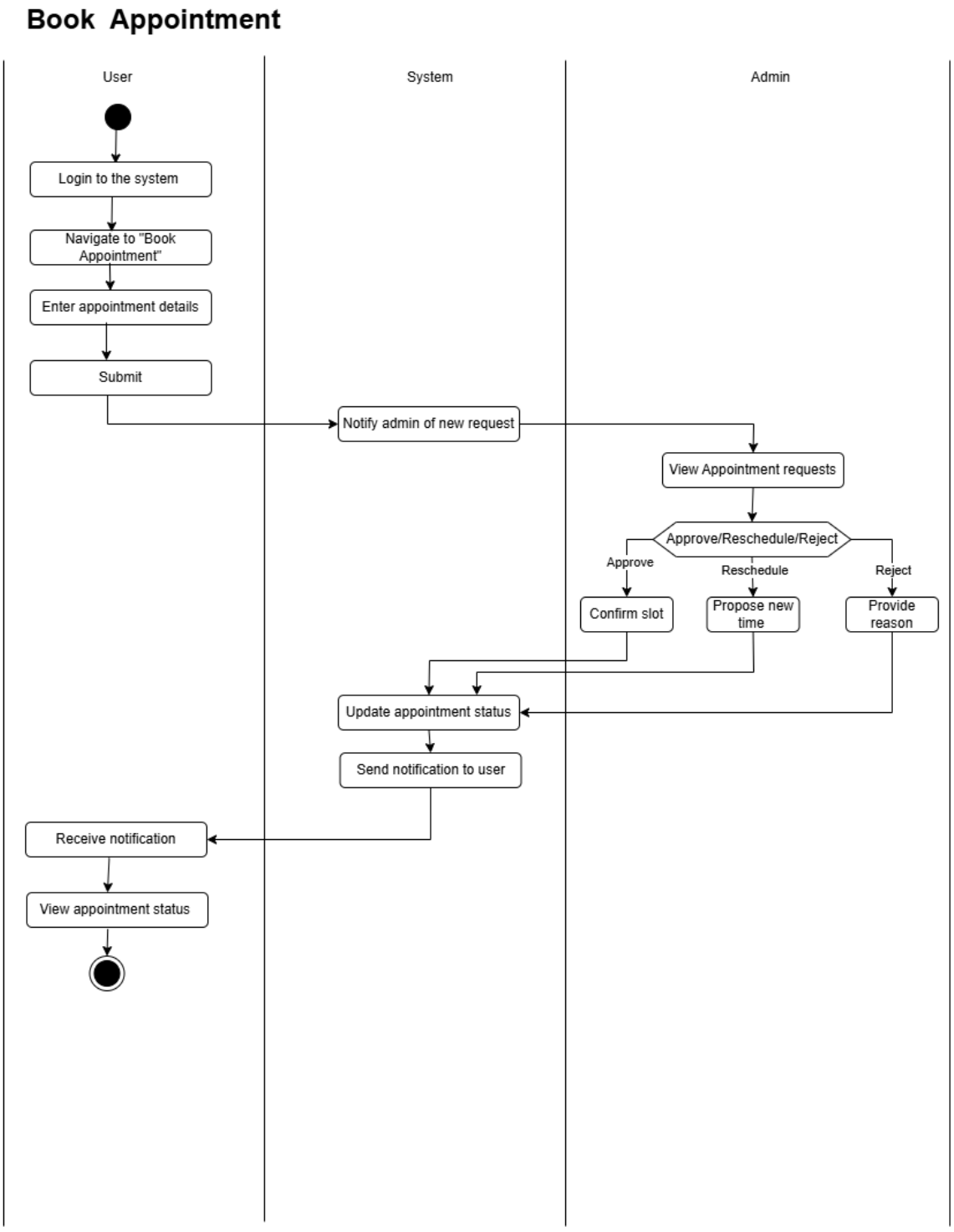


Figure 11.4: Activity Diagram — Book appointment

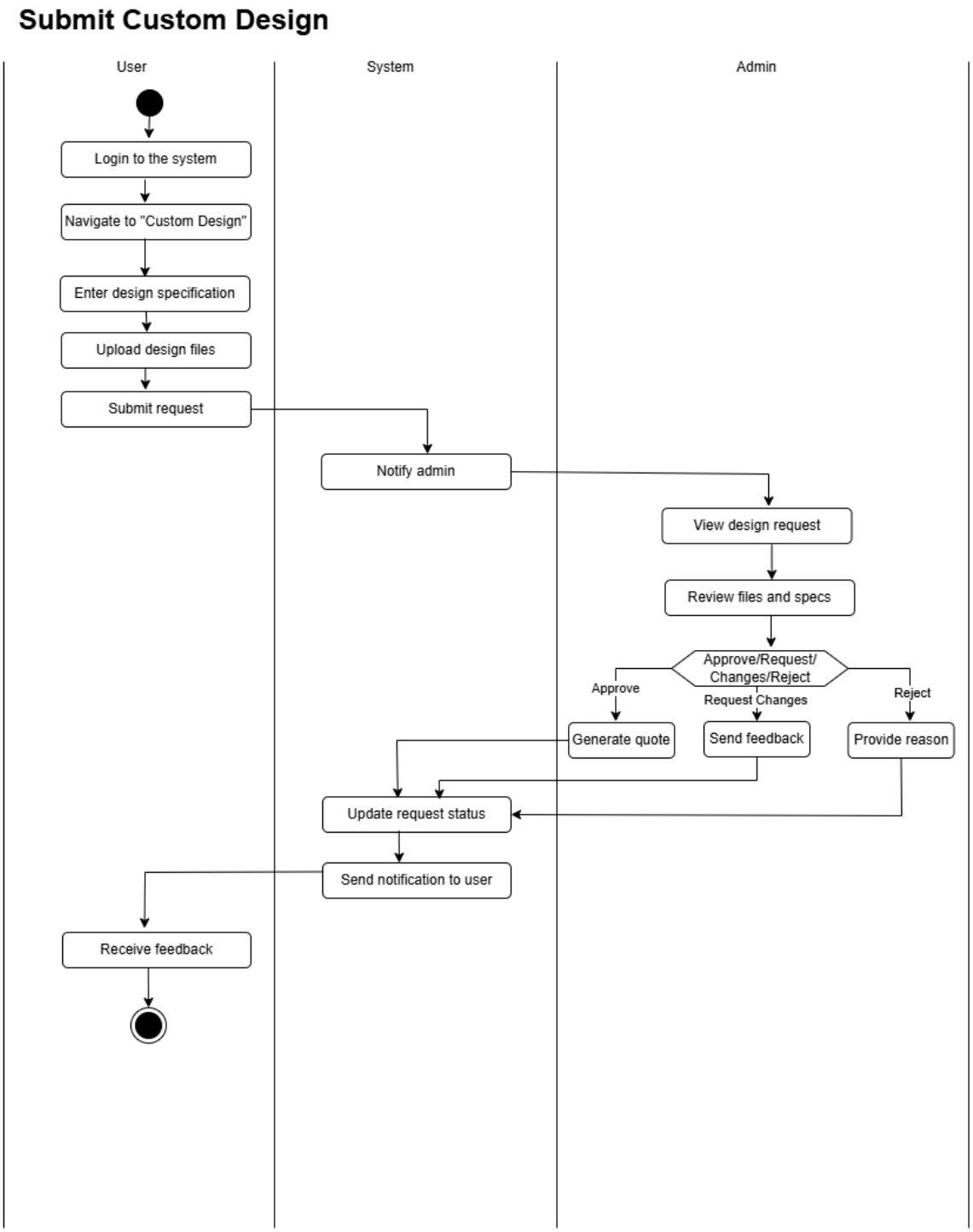


Figure 11.5: Activity Diagram — Submit custom design

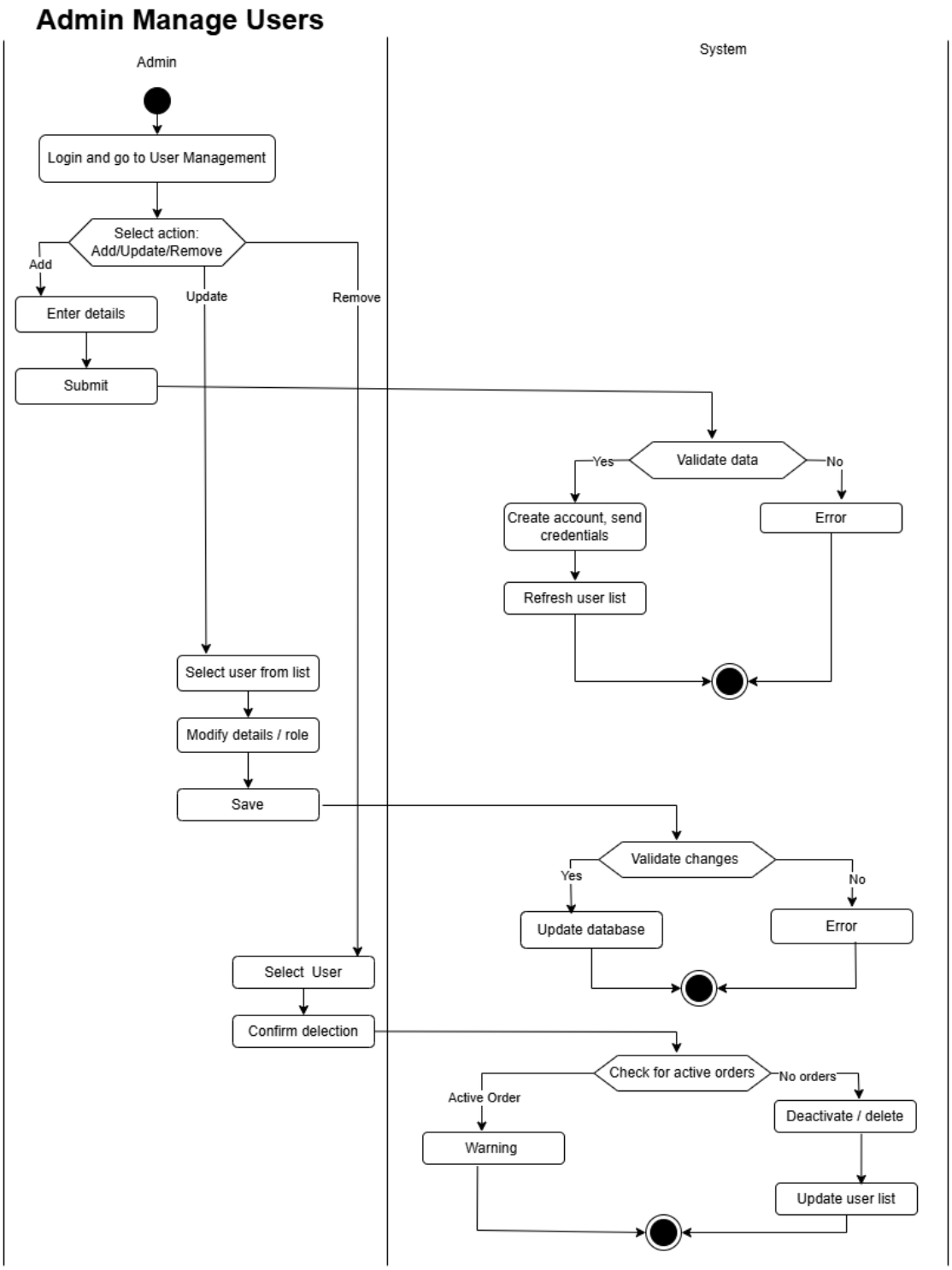


Figure 11.6: Activity Diagram — Admin mange users

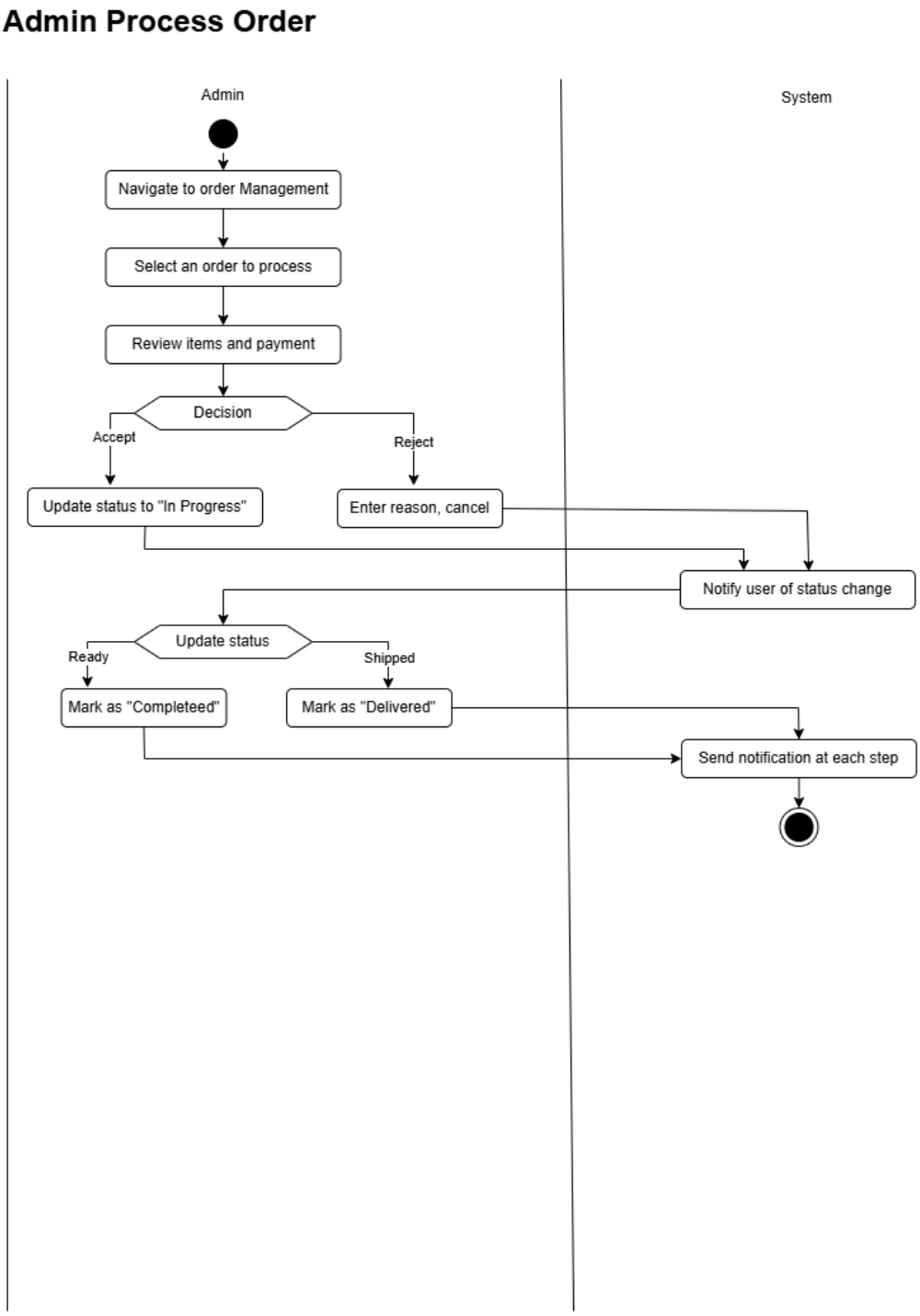


Figure 11.7: Activity Diagram — Admin Process Order

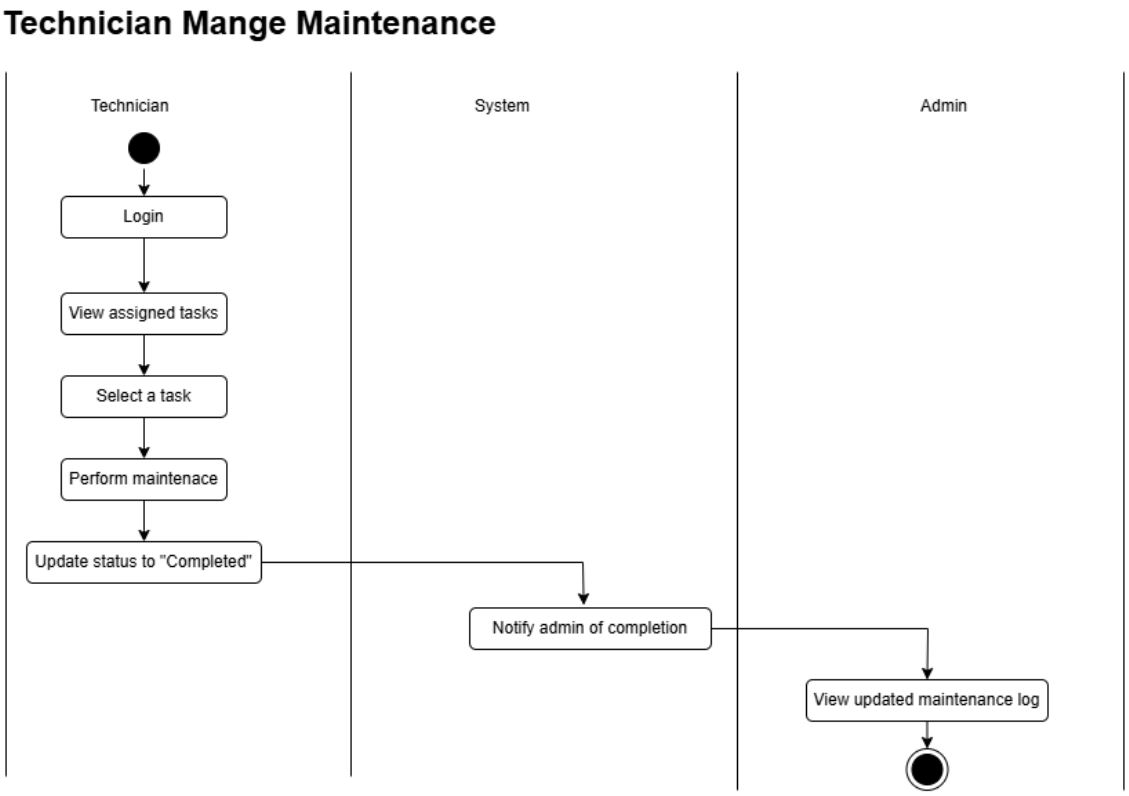


Figure 11.8: Activity Diagram — Technician manage maintenance

CHAPTER 12 EXTERNAL INTERFACE REQUIREMENTS

12.1.User interfaces

• Role-based dashboards can be displayed for each user type (Customer, Admin), including relevant information such as orders, bookings, payments, and system activities.

• Intuitive navigation can be provided for browsing machines, wood designs, and services through clearly structured menus and categories.

• Customers can be allowed to place orders, submit custom design requests, book appointments, and track order status through well-designed forms and pages.

• An admin dashboard can be provided with easy access to manage products, users, orders, payments, and maintenance schedules through clearly labeled options.

• Clear buttons, tabs, and navigation components can be used to ensure a simple and efficient user experience.

12.2.Hardware interfaces

• Support for iOS, Android, Windows, and macOS devices can be provided for both regular users and group administrators.

• Accurate tracking and engagement across all devices, including laptops, desktop computers, tablets, and cellphones can be ensured by the system.

12.3.Software interfaces

• Integration with a JWT-based authentication system can be implemented to provide secure user login and session management.

• MySQL database can be used to store user data, machine details, orders, payments, appointments, and maintenance records.

• Payment gateway APIs (such as Stripe or PayPal) can be integrated to handle secure online transactions.

• Cloud storage services (such as Cloudinary or AWS S3) can be used to upload and manage product images and custom design files.

• Email/SMS notification services can be integrated to send order confirmations, payment updates, and booking alerts to users.

12.4.Communication interfaces

• HTTPS protocol can be used for all communications between the client (frontend) and server (backend) to ensure secure data transmission.

• RESTful APIs can be provided for communication between the frontend (React) and backend (Spring Boot) to handle data such as users, products, orders, payments, and bookings.

• Real-time notifications can be sent using push notification services or in-app alerts to inform users about important updates such as order progress and approvals.

CHAPTER 13: DATA REQUIREMENTS

Table 13.1: Data requirements

Entity	Key Attributes
User	UserID, Name, Email, Password (hashed), Role (Admin/Customer), Phone, Address, CreatedAt
Product / Machine	ProductID, Name, Category, Description, Specifications, Price, StockQty, Images, ManualURL, CreatedAt
Order	OrderID, UserID, ProductID,OrderType (B2B/B2C), Items, TotalAmount, Status, PaymentMode, InvoiceRef, CreatedAt
Custom Design Request	RequestID, UserID, Description, Status, CreatedAt
Payment	PaymentID, OrderID, Amount, Method, StripeTransactionID, Status, PaidAt
Maintenance Record	MaintenanceID, MachineID, Type, ScheduledDate, Status, CompletedAt
Appointment	AppointmentID, UserID, RequestID,Date,Time, Purpose, Status (Pending/Approved/Rescheduled)

Notification	NotificationID, UserID, Type, Message, Status, CreatedAt
Feedback	FeedbackID, UserID, ProductID,OrderID, RequestID,Rating (1-5), Comment, SubmittedAt

CHAPTER 14: ALTERNATIVE SOLUTIONS CONSIDERED

14.1. Existing e-commerce platforms / frameworks

• Using platforms like Magento, WooCommerce, or Shopify for order management, payment processing, and product catalog features.

•Advantages: Ready-made features, secure payment integration, less development time.

•Disadvantages: Limited customization for custom wood design requests and maintenance scheduling.

14.2. Mobile application only

•Developing only a mobile application for managing orders, bookings, and user interactions.

• Advantages: Easy access via smartphones, supports on-the-go usage.

• Disadvantages: Not suitable for admin tasks that require desktops,higher development and maintenance complexity.

14.3. Spreadsheet-based management (excel / google sheets)

• Using spreadsheets to manually manage customer data, orders, and payments.

• Advantages: Low cost and easy to use, familiar interface for most users.

• Disadvantages: Not scalable for large operations,lacks real-time tracking and automation features.

14.4. Third-party notification services (firebase / twilio)

• Using external services to send SMS, email, or push notifications for system updates.

• Advantages: Enables real-time communication, improves user engagement.

• Disadvantages: Additional integration complexity, cost increases with usage.

14.5. Database alternatives (MongoDB vs MySQL)

• Using NoSQL databases like MongoDB instead of relational databases.

• Advantages: Flexible schema design, suitable for large and unstructured data.

• Disadvantages: Not ideal for relational data like orders and payments, complex query handling compared to MySQL.

14.6. Cloud-based storage solutions (AWS S3 / Cloudinary)

• Using cloud storage services to store product images and custom design files.

• Advantages: Scalable and reliable storage, fast access and content delivery.

• Disadvantages: Ongoing service costs,dependency on external cloud providers.

CHAPTER 15: FEASIBILITY STUDY

15.1. Technical feasibility

The technical feasibility of the proposed GMC Lanka Machine and Product Ordering System is evaluated by analyzing the technologies, tools, and alternative solutions that can be used to develop and deploy the system effectively.

15.1.1 Proposed system (GMC Lanka Solution)

The proposed system is technically feasible using modern web development technologies. The frontend will be developed using React.js, which provides a responsive and interactive user interface. The backend will be implemented using Node.js with Express.js, enabling efficient handling of application logic, API communication, and system operations.

A MySQL database will be used to manage structured data such as user information, product details, orders, payments, and maintenance records. Secure authentication will be implemented using JWT (JSON Web Tokens) to ensure safe user login and session management.

Integration with the Stripe payment gateway will be implemented to support secure online transactions.Additionally, cloud services such as AWS and Cloudinary will be utilized for hosting and media storage.

The architecture is designed to be scalable and capable of supporting multiple users simultaneously without performance degradation. However, careful implementation is required to ensure system security, payment reliability, and performance optimization.

15.1.2 Existing eCommerce platforms (Magento / WooCommerce /Shopify)

The use of existing eCommerce platforms such as Magento, WooCommerce, or Shopify is also considered technically feasible.Ready-made functionalities, including product management, order processing, and secure payment integration, are provided by these platforms and development time can be significantly reduced.

However, these platforms have limitations in terms of customization. Specifically, features such as custom machine design requests, maintenance scheduling, and technician management are not natively supported. Additional technical complexity may be introduced when these platforms are extended to meet the specific requirements of GMC Lanka.

15.1.3 Cloud-based infrastructure

The system can be deployed using a cloud-based infrastructure, which is technically feasible and highly scalable. Reliable services for hosting, database management, storage, and system monitoring can be provided by Cloud platforms such as AWS

High availability, data backup, and easy integration with third-party services such as payment gateways and notification systems can be ensured by Cloud deployment. However, proper configuration and cost management are necessary to maintain efficiency and avoid unnecessary expenses.

15.1.4 Manual / Existing system

The current manual or semi-digital system is considered technically simple but inefficient.Features such as real-time tracking, automated billing, and centralized data management are lacking. While minimal technical resources are required, the system is not considered suitable for scaling business operations or improving customer experience. Therefore, the transition to a fully digital system is technically justified and considered beneficial.

15.2.Economic feasibility

The economic feasibility of the GMC Lanka Machine and Product Ordering System evaluates the cost-effectiveness of developing and implementing the proposed system.

15.2.1 Development costs

The system will be developed as part of an academic project; therefore, no direct labor costs are incurred for development. However, indirect costs may include internet usage, development tools, and optional premium services.

The technologies used, such as React.js, Node.js, and MySQL, are open-source and free to use, which significantly reduces software costs.

15.2.2 Infrastructure costs

The system will be deployed using cloud services such as AWS. Initial deployment may use free-tier services, but long-term usage will require payment based on resource consumption (hosting, storage, and bandwidth).

Cloudinary or similar services may also incur costs for storing product images and media files beyond free limits.

15.2.3 Operational costs

Ongoing costs include:

- Cloud hosting and database maintenance
- Payment gateway transaction fees (Stripe charges per transaction)
- System maintenance and updates

These costs are manageable and scale according to system usage.

15.2.4 Cost-benefit analysis

Although there are some operational costs, the long-term benefits outweigh the expenses. The system improves efficiency, reduces errors, and supports business growth, making it economically feasible.

15.3.Operational feasibility

Operational feasibility can be evaluated by how well the proposed system will function within the organization and how easily users can adopt it.

15.3.1 User acceptance

The system is designed with a user-friendly web interface, making it easy for customers, administrators, and technicians to use. Since most users are familiar with web applications, minimal training is required.

15.3.2 Ease of use

The interface will be simple and intuitive, allowing users to:

- Browse products and place orders
- Track order status
- Manage appointments and maintenance schedules

Admin and technician dashboards will provide clear functionalities for managing operations.

15.3.3 Organizational fit

The system aligns with the business processes of GMC Lanka by digitalizing machine ordering, customer communication, and service management. It replaces manual processes with an integrated platform, improving overall workflow efficiency.

15.3.4 Technical support and maintenance

The system can be maintained by individuals with basic technical knowledge of web applications. Documentation and support will be provided to ensure smooth operation.

15.4.Schedule feasibility

Schedule feasibility evaluates whether the project can be completed within the given time frame.

15.4.1 Project timeline

The development of the GMC Lanka system is planned within the academic project duration. The project is divided into phases such as:

- Requirement gathering and analysis
- System design
- Development (frontend and backend)
- Testing and debugging
- Deployment and documentation

15.4.2 Resource availability

The project is carried out by a team of students with knowledge of modern web technologies. Required tools and technologies are readily available, which supports timely development.

15.4.3 Risk factors

Potential risks that may affect the schedule include:

- Technical challenges during development
- Integration issues with payment gateway or cloud services
- Time constraints due to academic workload

15.4.4 Mitigation strategies

To manage risks, the following strategies will be used:

- Proper planning and task allocation
- Regular progress monitoring
- Early testing and debugging
- Using reliable frameworks and tools

CHAPTER 16: RISK ANALYSIS

16.1 Risk analysis

Table 16.1:Risk register

Risk ID	Description	Likelihood	Impact	Risk Level	Mitigation Strategy
R001	Integration issues with Stripe payment gateway	Medium	High	High	Use official Stripe API, perform thorough testing, and implement fallback handling

					for failed transactions
R002	Performance issues with large number of users	Medium	High	High	Optimize database queries, use scalable cloud infrastructure (AWS), and conduct load testing
R003	Delay in project completion due to academic workload	High	Medium	High	Create proper schedule, divide tasks among team members, and track progress regularly
R004	Lack of experience with new technologies (React, Node.js)	Medium	Medium	Medium	Conduct self-learning, use online resources, and practice before implementation
R005	Requirement changes during development	Medium	Medium	Medium	Maintain clear documentation and communicate regularly with stakeholders
R006	Data security vulnerabilities (user data, payments)	Low	High	Medium	Implement JWT authentication, encryption, and secure coding practices
R007	Cloud service downtime	Low	High	Medium	Use reliable cloud providers and

	(AWS/Cloudinary)				implement backup/recovery mechanisms
R008	Internet connectivity issues for users	Medium	Medium	Medium	Optimize system performance and allow retry mechanisms
R009	Incorrect data entry by users/admin	Medium	Low	Low	Implement input validation and user-friendly forms
R010	Limited customization in third-party integrations	Low	Medium	Low	Use flexible APIs and custom backend logic

16.2.Risk categories

16.2.1.Technical risks

Risks related to system development, integration, and performance.
Eg: Payment gateway integration issues, system performance, security vulnerabilities.

16.2.2.Schedule risks

Risks that may delay project completion.
Eg: Academic workload, unexpected technical challenges, delays in testing.

16.2.3.Resource risks

Risks related to availability of team members or technical skills.
Eg: Lack of experience with frameworks, limited development time.

16.2.4.Requirements risks

Risks arising from unclear or changing requirements.
Eg: Requirement changes, misunderstanding of client needs.

16.2.5.External / Organisational risks

Risks caused by external systems or organizational factors.
Eg: Cloud service downtime, internet issues, dependency on third-party services like Stripe.

CHAPTER 17: REFERENCES

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Client Representative Sign-off

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Title: Founder & CEO

Organization: Grand Machine Carving Lanka (Pvt) Ltd

Date: 2026/03/21

Signature: _____

Project Supervisor Sign-off

Name: _____

Date: _____

Signature: _____